

Program Analysis 2025-26

Lecture #18

Control Flow Analysis



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Exercises

Exercises

Consider and add labels to the following expressions
Guess analysis results and verify them

$$e \triangleq ((\text{fn } x \Rightarrow (x + 1)) (3))$$

$$e \triangleq ((\text{fn } f \Rightarrow ((f \ 76) + (f \ 77))) (\text{fn } a \Rightarrow a))$$

$$e \triangleq \text{let } f = \text{fn } x \Rightarrow x + 1 \\ \text{in let } g = \text{fn } y \Rightarrow y + 2 \\ \text{in let } h = \text{fn } z \Rightarrow z + 3 \\ \text{in } (f \ g) + (f \ h)$$

Exercise 1

$$[app] (\hat{C}, \hat{\rho}) \models (t_1^{\ell_1} t_2^{\ell_2})^{\ell} \text{ iff } (\hat{C}, \hat{\rho}) \models t_1^{\ell_1} \wedge (\hat{C}, \hat{\rho}) \models t_2^{\ell_2} \\ (\forall (\text{fn } x \Rightarrow t_0^{\ell_0}) \in \hat{C}(\ell_1) : \\ (\hat{C}, \hat{\rho}) \models t_0^{\ell_0} \wedge \\ \hat{C}(\ell_2) \subseteq \hat{\rho}(x) \wedge \hat{C}(\ell_0) \subseteq \hat{C}(\ell)) \wedge$$

Consider and add labels to the following expression
Guess an analysis result and verify it

$$e \triangleq ((\text{fn } x \Rightarrow (x^1 + 1^2)^3)^4 (3)^5)^6$$

$\hat{C}(1)$	$\emptyset$ (3)
$\hat{C}(2)$	$\emptyset$ (1)
$\hat{C}(3)$	$\emptyset$ (4)
$\hat{C}(4)$	$\{\text{fn } x \Rightarrow (x^1 + 1^2)^3\}$
$\hat{C}(5)$	$\emptyset$ (3)
$\hat{C}(6)$	$\emptyset$ (4)
$\hat{\rho}(x)$	$\emptyset$ (3)

Exercise 1

$$[app] (\hat{C}, \hat{\rho}) \models (t_1^{\ell_1} t_2^{\ell_2})^{\ell} \text{ iff } (\hat{C}, \hat{\rho}) \models t_1^{\ell_1} \wedge (\hat{C}, \hat{\rho}) \models t_2^{\ell_2}$$

$$(\forall (\text{fn } x \Rightarrow t_0^{\ell_0}) \in \hat{C}(\ell_1) : (\hat{C}, \hat{\rho}) \models t_0^{\ell_0} \wedge \hat{C}(\ell_2) \subseteq \hat{\rho}(x) \wedge \hat{C}(\ell_0) \subseteq \hat{C}(\ell)) \wedge$$

Consider and add labels to the following expression
Guess an analysis result and verify it

$$e \triangleq ((\text{fn } x \Rightarrow (x^1 + 1^2)^3)^4 (3)^5)^6$$

$$(\hat{C}, \hat{\rho}) \models e \text{ iff } (\hat{C}, \hat{\rho}) \models (\text{fn } x \Rightarrow (x^1 + 1^2)^3)^4 \text{ iff } (\hat{C}, \hat{\rho}) \models \{\text{fn } x \Rightarrow (x^1 + 1^2)^3\} \subseteq \hat{C}(4)$$

$$\wedge (\hat{C}, \hat{\rho}) \models (3)^5 \text{ iff } \emptyset \subseteq \hat{C}(5)$$

$$\{\text{fn } x \Rightarrow (x^1 + 1^2)^3\} \subseteq \hat{C}(4) \Rightarrow (\hat{C}, \hat{\rho}) \models (x^1 + 1^2)^3 \wedge$$

$$\hat{C}(5) \subseteq \hat{\rho}(x) \wedge \hat{C}(3) \subseteq \hat{C}(6)$$

$$(\hat{C}, \hat{\rho}) \models (x^1 + 1^2)^3 \text{ iff } (\hat{C}, \hat{\rho}) \models x^1 \text{ iff } \hat{\rho}(x) \subseteq \hat{C}(1)$$

$$\wedge (\hat{C}, \hat{\rho}) \models 1^2 \text{ iff } \emptyset \subseteq \hat{C}(2)$$

Exercise 1

$$[app] (\hat{C}, \hat{\rho}) \models (t_1^{\ell_1} t_2^{\ell_2})^{\ell} \text{ iff } (\hat{C}, \hat{\rho}) \models t_1^{\ell_1} \wedge (\hat{C}, \hat{\rho}) \models t_2^{\ell_2}$$

$$(\forall (\text{fn } x \Rightarrow t_0^{\ell_0}) \in \hat{C}(\ell_1) : (\hat{C}, \hat{\rho}) \models t_0^{\ell_0} \wedge \hat{C}(\ell_2) \subseteq \hat{\rho}(x) \wedge \hat{C}(\ell_0) \subseteq \hat{C}(\ell)) \wedge$$

Consider and add labels to the following expression
Guess an analysis result and verify it

$$e \triangleq ((\text{fn } x \Rightarrow (x^1 + 1^{23})^4) (3)^{56})^6$$

$$\{\text{fn } x \Rightarrow (x^1 + 1^{23})\} \subseteq \hat{C}(4) \quad \checkmark$$

$$\emptyset \subseteq \hat{C}(5) \quad \checkmark$$

$$\hat{C}(5) \subseteq \hat{\rho}(x) \quad \checkmark$$

$$\hat{C}(3) \subseteq \hat{C}(6) \quad \checkmark$$

$$\hat{\rho}(x) \subseteq \hat{C}(1) \quad \checkmark$$

$$\emptyset \subseteq \hat{C}(2) \quad \checkmark$$

$\hat{C}(1)$	$\emptyset$ (3)
$\hat{C}(2)$	$\emptyset$ (1)
$\hat{C}(3)$	$\emptyset$ (4)
$\hat{C}(4)$	$\{\text{fn } x \Rightarrow (x^1 + 1^{23})\}$
$\hat{C}(5)$	$\emptyset$ (3)
$\hat{C}(6)$	$\emptyset$ (4)
$\hat{\rho}(x)$	$\emptyset$ (3)



Exercise 2

Consider and add labels to the following expression: guess an analysis result and verify it

$$e \triangleq ((\text{fn } f \Rightarrow ((f^{1} 76^{2})^{3} + (f^{4} 77^{5})^{6})^{7})^{8} (\text{fn } a \Rightarrow a^{9})^{10})^{12}$$

Exercise 2

$$e \triangleq ((\text{fn } f \Rightarrow ((f^1 76^2)^3 + (f^4 77^5)^6)^7)^8 (\text{fn } a \Rightarrow a^9)^{10})^{12}$$

$$\hat{C}(1) = \{\text{fn } a \Rightarrow a^9\}$$

$$\hat{C}(2) = \emptyset \text{ (76)}$$

$$\hat{C}(3) = \emptyset \text{ (76)}$$

$$\hat{C}(4) = \{\text{fn } a \Rightarrow a^9\}$$

$$\hat{C}(5) = \emptyset \text{ (77)}$$

$$\hat{C}(6) = \emptyset \text{ (77)}$$

$$\hat{C}(7) = \emptyset \text{ (76+77)}$$

$$\hat{C}(8) = \{\text{fn } f \Rightarrow ((f^1 76^2)^3 + (f^4 77^5)^6)^2\}$$

$$\hat{C}(9) = \emptyset \text{ (76, 77)}$$

$$\hat{C}(10) = \{\text{fn } a \Rightarrow a^9\}$$

$$\hat{C}(11) = \emptyset$$

$$\hat{C}(12) = \{\text{fn } f \Rightarrow ((f^1 76^2)^3 + (f^4 77^5)^6)^2\}$$

$$\hat{\rho}(f) = \{\text{fn } a \Rightarrow a^9\}$$

$$\hat{\rho}(a) = \emptyset \text{ (76, 77)}$$

Exercise 2

[app] $(\hat{C}, \hat{\rho}) \models (t_1^{\ell_1} t_2^{\ell_2})^\ell$ iff $(\hat{C}, \hat{\rho}) \models t_1^{\ell_1} \wedge (\hat{C}, \hat{\rho}) \models t_2^{\ell_2}$
 $(\forall (\text{fn } x \Rightarrow t_0^{\ell_0}) \in \hat{C}(\ell_1) :$
 $(\hat{C}, \hat{\rho}) \models t_0^{\ell_0} \wedge$
 $\hat{C}(\ell_2) \subseteq \hat{\rho}(x) \wedge \hat{C}(\ell_0) \subseteq \hat{C}(\ell)) \wedge$

$$e \triangleq ((\text{fn } f \Rightarrow ((f^1 76^2)^3 + (f^4 77^5)^6)^7)^8 (\text{fn } a \Rightarrow a^9)^{10})^{12}$$

$$\begin{aligned} (\hat{C}, \hat{\rho}) \models e &\text{ iff } (\text{fn } f \Rightarrow ((f^1 76^2)^3 + (f^4 77^5)^6)^7)^8 \text{ iff } \{\text{fn } f \Rightarrow ((f^1 76^2)^3 + (f^4 77^5)^6)^7\} \in \hat{C}(8) \\ &\wedge (\hat{C}, \hat{\rho}) \models (\text{fn } a \Rightarrow a^9)^{10} \text{ iff } \{\text{fn } a \Rightarrow a^9\} \in \hat{C}(10) \wedge \\ &\wedge (\hat{C}, \hat{\rho}) \models ((f^1 76^2)^3 + (f^4 77^5)^6)^7 \text{ iff } (\hat{C}, \hat{\rho}) \models (f^1 76^2)^3 \wedge (\hat{C}, \hat{\rho}) \models (f^4 77^5)^6 \\ &\wedge \hat{C}(10) \subseteq \hat{\rho}(f) \wedge \hat{C}(8) \subseteq \hat{C}(12) \end{aligned}$$

$$\begin{aligned} (\hat{C}, \hat{\rho}) \models (f^1 76^2)^3 &\text{ iff } (\hat{C}, \hat{\rho}) \models f^1 \wedge (\hat{C}, \hat{\rho}) \models 76^2 \wedge (\hat{C}, \hat{\rho}) \models a^9 \wedge \hat{C}(2) \subseteq \hat{\rho}(a) \wedge \hat{C}(9) \subseteq \hat{C}(3) \\ (\hat{C}, \hat{\rho}) \models (f^4 77^5)^6 &\text{ iff } (\hat{C}, \hat{\rho}) \models f^4 \wedge (\hat{C}, \hat{\rho}) \models 77^5 \wedge (\hat{C}, \hat{\rho}) \models a^9 \wedge \hat{C}(5) \subseteq \hat{\rho}(a) \wedge \hat{C}(9) \subseteq \hat{C}(6) \\ (\hat{C}, \hat{\rho}) \models f^1 &\text{ iff } \hat{\rho}(f) \subseteq \hat{C}(1) \wedge (\hat{C}, \hat{\rho}) \models a^9 \text{ iff } \hat{\rho}(a) \subseteq \hat{C}(9) \\ (\hat{C}, \hat{\rho}) \models f^4 &\text{ iff } \hat{\rho}(f) \subseteq \hat{C}(4) \wedge \end{aligned}$$

Exercise 2

$((\text{fn } f \Rightarrow ((f^1 76^{23}) + (f^4 77^{56})^7)^8 (\text{fn } a \Rightarrow a^9)^{10})^{12}$

$\{\text{fn } f \Rightarrow ((f^1 76^{23}) + (f^4 77^{56})^2)\} \in \hat{C}(8)$

$\{\text{fn } a \Rightarrow a^9\} \in \hat{C}(10)$

$\hat{C}(10) \subseteq \hat{\rho}(f)$

$\hat{C}(8) \subseteq \hat{C}(12)$

$\hat{C}(2) \subseteq \hat{\rho}(a)$

$\hat{C}(9) \subseteq \hat{C}(3)$

$\hat{C}(9) \subseteq \hat{C}(6)$

$\hat{\rho}(f) \subseteq \hat{C}(1)$

$\hat{\rho}(f) \subseteq \hat{C}(4)$

$\hat{\rho}(a) \subseteq \hat{C}(9)$

$\hat{C}(1) = \{\text{fn } a \Rightarrow a^9\}$

$\hat{C}(2) = \emptyset$ (76)

$\hat{C}(3) = \emptyset$ (76)

$\hat{C}(4) = \{\text{fn } a \Rightarrow a^9\}$

$\hat{C}(5) = \emptyset$ (77)

$\hat{C}(6) = \emptyset$ (77)

$\hat{C}(7) = \emptyset$ (76+77)

$\hat{C}(8) = \{\text{fn } f \Rightarrow ((f^1 76^{23}) + (f^4 77^{56})^2)\}$

$\hat{C}(9) = \emptyset$ (76, 77)

$\hat{C}(10) = \{\text{fn } a \Rightarrow a^9\}$

$\hat{C}(11) = \emptyset$

$\hat{C}(12) = \{\text{fn } f \Rightarrow ((f^1 76^{23}) + (f^4 77^{56})^2)\}$

$\hat{\rho}(f) = \{\text{fn } a \Rightarrow a^9\}$

$\hat{\rho}(a) = \emptyset$ (76, 77)



Exercise 3

Consider and add labels to the following expression

Guess an analysis result and verify it

```
let f = fn x => x 1
  in let g = fn y => y + 2
    in let h = fn z => z + 3
      in (f g) + (f h)
```

$\text{id}_w \triangleq \text{fn } w \Rightarrow w^{\ell}$

$e \triangleq$ (let f = (fn x => (x¹ 1²)³)⁴
 in (let g = (fn y => (y⁵ + 2⁶)⁷)⁸
 in (let h = (fn z => (z⁹ + 3¹⁰)¹¹)¹²
 in ((f¹³ g¹⁴)¹⁵ + (f¹⁶ h¹⁷)¹⁸)¹⁹)²⁰)²¹)²²

Theoretical properties

Semantic correctness

Goal: show that acceptable CFA results are semantically correct

The analysis information is a safe description of what will happen during the evaluation of the program, i.e., a **sound over-approximation** of the runtime behaviour

To reason on correctness, we define a **Small Step Structural Operational Semantics** for FUN, where:

- We use **intermediate expressions** *ie* to handle the binding of local variables
- We choose an approach based on **explicit environments** (matching the CFA abstraction) rather than substitutions

Subject reduction result

If the abstract ($\hat{\rho}$) and concrete environments (ρ) agree ($\rho \mathcal{R} \hat{\rho}$):

$(\hat{C}, \hat{\rho}) \models ie$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie'$

to preserve the identity of functions (and hence abstract values) during evaluation

$\rho \mathcal{R} \hat{\rho}$ iff $dom(\rho) \subseteq dom(\hat{\rho}) \wedge \forall x \in dom(\rho) \forall t_x \forall \rho_x :$
 $(\rho(x) = \text{close } t_x \text{ in } \rho_x) \Rightarrow (t_x \in \hat{\rho}(x) \wedge \rho_x \mathcal{R} \hat{\rho})$

Structural Operational Semantic

Configurations

$v \in \mathbf{Val}$: values

$\rho \in \mathbf{Env}$: environments (mapping free variables to their *values*)

$v ::= c \mid \text{close } t \text{ in } \rho$

$\rho ::= [] \mid \rho[x \mapsto v]$

$ie ::= it^\ell$

Functions evaluate to
closures capturing
their environment

$ie \in \mathbf{IExp}$: intermediate expressions

$it \in \mathbf{ITerm}$: intermediate terms

$t ::= c \mid x \mid \text{fn } x \Rightarrow e_0 \mid \text{fun } f \ x \Rightarrow e_0$

$\mid ie_1 \ ie_2 \mid \text{if } ie_0 \text{ then } e_1 \text{ else } e_2 \mid \text{let } x = ie_1 \text{ in } e_2 \mid ie_1 \text{ op } ie_2$

$\mid \text{bind } \rho \text{ in } ie \text{ // bind the environment (encoding of frames)}$

$\mid \text{close } t \text{ in } \rho \text{ // closure}$

Transitions: $\rho \vdash ie \rightarrow ie'$

Rule for variables

$$[var] \rho \vdash x^\ell \rightarrow v^\ell \text{ iff } x \in Dom(\rho) \text{ and } \rho(x) = v$$

The value of a variable is obtained from the environment

Example

$$x^1 \rightarrow 7 \text{ if } \rho(x) = 7$$

Rule for the let binding

$$[let_2] \ \rho \vdash (\text{let } x = v_1^{\ell_1} \text{ in } e_2)^{\ell} \rightarrow \text{bind } (\rho_0[x \mapsto v_1] \text{ in } e_2)^{\ell} \\ \text{where } \rho_0 = \rho \upharpoonright FV(e_2)$$

Example

$$e \triangleq (\text{let } f = (\text{fn } x \Rightarrow x^1)^2 \text{ in } (f^3 \ 3^4)^5)$$

$$\rho = [g \mapsto (\text{fn } y \Rightarrow y^1)^2] \text{ and } \rho_0 = \rho \upharpoonright \{f\} = []$$

Rule for functions

$$[fn] \rho \vdash (fn\ x \Rightarrow e_0)^\ell \rightarrow (close\ (fn\ x \Rightarrow e_0)\ in\ \rho)^\ell \rightarrow$$

where $\rho_0 = \rho \mid FV(fn\ x \Rightarrow e_0)$

This axiom constructs the appropriate closure

It restricts the environment ρ to the free variables of the abstraction

Note that the label ℓ is preserved in the closure

Example

$$e \triangleq (fn\ x \Rightarrow x^1\ y^2)^3$$

$$\rho_0 = \rho \mid \{y\} \text{ since } FV(fn\ x \Rightarrow x^1\ y^2) = y$$

Rule for function application

$$[app_{fn}] \rho \vdash ((\text{close } (\text{fn } x \Rightarrow e_1) \text{ in } \rho_1)^{\ell_1} v_2^{\ell_2})^{\ell} \rightarrow (\text{bind } \rho_1[x \mapsto v_2] \text{ in } e_1)^{\ell}$$

This axiom describes the binding of the actual parameter to the formal parameter

Example

$$[y \mapsto 5] \vdash ((\text{close } (\text{fn } x \Rightarrow x^1 + y^2)^3 \text{ in } [y \mapsto 7])^4 3^5)^6 \rightarrow (\text{bind } [y \mapsto 7, x \mapsto 3] \text{ in } x^1 + y^2)^6$$

we use the definition environment, not the call one

Structural Operational Semantic I

$$\begin{array}{ll}
 [\text{var}] & \rho \vdash x^\ell \rightarrow v^\ell \text{ if } x \in \text{dom}(\rho) \text{ and } v = \rho(x) \\
 [\text{fn}] & \rho \vdash (\text{fn } x \Rightarrow e_0)^\ell \rightarrow (\text{close } (\text{fn } x \Rightarrow e_0) \text{ in } \rho_0)^\ell \\
 & \text{where } \rho_0 = \rho \mid FV(\text{fn } x \Rightarrow e_0) \\
 [\text{fun}] & \rho \vdash (\text{fun } f \ x \Rightarrow e_0)^\ell \rightarrow (\text{close } (\text{fun } f \ x \Rightarrow e_0) \text{ in } \rho_0)^\ell \\
 & \text{where } \rho_0 = \rho \mid FV(\text{fun } f \ x \Rightarrow e_0) \\
 [\text{app}_1] & \frac{\rho \vdash ie_1 \rightarrow ie'_1}{\rho \vdash (ie_1 \ ie_2)^\ell \rightarrow (ie'_1 \ ie_2)^\ell} \\
 [\text{app}_2] & \frac{\rho \vdash ie_2 \rightarrow ie'_2}{\rho \vdash (v_1^{\ell_1} \ ie_2)^\ell \rightarrow (v_1^{\ell_1} \ ie'_2)^\ell} \\
 [\text{app}_{fn}] & \rho \vdash ((\text{close } (\text{fn } x \Rightarrow e_1) \text{ in } \rho_1)^{\ell_1} \ v_2^{\ell_2})^\ell \rightarrow \\
 & \quad (\text{bind } \rho_1[x \mapsto v_2] \text{ in } e_1)^\ell \\
 [\text{app}_{fun}] & \rho \vdash ((\text{close } (\text{fun } f \ x \Rightarrow e_1) \text{ in } \rho_1)^{\ell_1} \ v_2^{\ell_2})^\ell \rightarrow \\
 & \quad (\text{bind } \rho_2[x \mapsto v_2] \text{ in } e_1)^\ell \\
 & \quad \text{where } \rho_2 = \rho_1[f \mapsto \text{close } (\text{fun } f \ x \Rightarrow e_1) \text{ in } \rho_1] \\
 [\text{bind}_1] & \frac{\rho_1 \vdash ie_1 \rightarrow ie'_1}{\rho \vdash (\text{bind } \rho_1 \text{ in } ie_1)^\ell \rightarrow (\text{bind } \rho_1 \text{ in } ie'_1)^\ell} \\
 [\text{bind}_2] & \rho \vdash (\text{bind } \rho_1 \text{ in } v_1^{\ell_1})^\ell \rightarrow v_1^\ell
 \end{array}$$

Structural Operational Semantic II

$$\begin{array}{l} [if_1] \quad \frac{\rho \vdash ie_0 \rightarrow ie'_0}{\rho \vdash (\text{if } ie_0 \text{ then } e_1 \text{ else } e_2)^\ell \rightarrow (\text{if } ie'_0 \text{ then } e_1 \text{ else } e_2)^\ell} \\ [if_2] \quad \rho \vdash (\text{if true}^{\ell_0} \text{ then } t_1^{\ell_1} \text{ else } t_2^{\ell_2})^\ell \rightarrow t_1^\ell \\ [if_3] \quad \rho \vdash (\text{if false}^{\ell_0} \text{ then } t_1^{\ell_1} \text{ else } t_2^{\ell_2})^\ell \rightarrow t_2^\ell \\ [let_1] \quad \frac{\rho \vdash ie_1 \rightarrow ie'_1}{\rho \vdash (\text{let } x = ie_1 \text{ in } e_2)^\ell \rightarrow (\text{let } x = ie'_1 \text{ in } e_2)^\ell} \\ [let_2] \quad \rho \vdash (\text{let } x = v^{\ell_1} \text{ in } e_2)^\ell \rightarrow (\text{bind } \rho_0[x \mapsto v] \text{ in } e_2)^\ell \\ \quad \text{where } \rho_0 = \rho \mid FV(e_2) \\ [op_1] \quad \frac{\rho \vdash ie_1 \rightarrow ie'_1}{\rho \vdash (ie_1 \text{ op } ie_2)^\ell \rightarrow (ie'_1 \text{ op } ie_2)^\ell} \\ [op_2] \quad \frac{\rho \vdash ie_2 \rightarrow ie'_2}{\rho \vdash (v_1^{\ell_1} \text{ op } ie_2)^\ell \rightarrow (v_1^{\ell_1} \text{ op } ie'_2)^\ell} \\ [op_3] \quad \rho \vdash (v_1^{\ell_1} \text{ op } v_2^{\ell_2})^\ell \rightarrow v^\ell \quad \text{if } v = v_1 \text{ op } v_2 \end{array}$$

Back to Example 1

$((\text{fn } x \Rightarrow x^1)^2 (\text{fn } y \Rightarrow y^3)^4)^5$

$[] \vdash ((\text{fn } x \Rightarrow x^1)^2 (\text{fn } y \Rightarrow y^3)^4)^5$
→ $((\text{close}(\text{fn } x \Rightarrow x^1) \text{ in } [])^2 (\text{fn } y \Rightarrow y^3)^4)^5$
→ $((\text{close}(\text{fn } x \Rightarrow x^1) \text{ in } [])^2 (\text{close } (\text{fn } y \Rightarrow y^3) \text{ in } [])^4)^5$
→ $(\text{bind } [x \mapsto (\text{close } (\text{fn } y \Rightarrow y^3) \text{ in } [])] \text{ in } x^1)^5$
→ $(\text{bind } [x \mapsto (\text{close } (\text{fn } y \Rightarrow y^3) \text{ in } [])] \text{ in } (\text{close } (\text{fn } y \Rightarrow y^3)^4 \text{ in } [])^1)^5$
→ $(\text{close } (\text{fn } y \Rightarrow y^3) \text{ in } [])^5$

New CFA clauses

$$\begin{aligned} [bind] \quad (\hat{C}, \hat{\rho}) \models (\text{bind } \rho \text{ in } it_0^{\ell_0})^\ell & \text{ iff } (\hat{C}, \hat{\rho}) \models it_0^{\ell_0} \wedge \\ & \hat{C}(\ell_0) \subseteq \hat{C}(\ell) \wedge \rho \mathcal{R} \hat{\rho} \\ [close] \quad (\hat{C}, \hat{\rho}) \models (\text{close } t_0 \text{ in } \rho)^\ell & \text{ iff } \{t_0\} \subseteq \hat{C}(\ell) \wedge \rho \mathcal{R} \hat{\rho} \end{aligned}$$

$$\begin{array}{l} \rho \mathcal{R} \hat{\rho} \text{ iff } \text{dom}(\rho) \subseteq \text{dom}(\hat{\rho}) \wedge \forall x \in \text{dom}(\rho) \forall t_x \forall \rho_x : \\ \quad (\rho(x) = \text{close } t_x \text{ in } \rho_x) \Rightarrow (t_x \in \hat{\rho}(x) \wedge \rho_x \mathcal{R} \hat{\rho}) \end{array}$$

For the other clauses just replace e with ie

Relation between the environments

$$\rho \mathcal{R} \hat{\rho} \text{ iff } \text{dom}(\rho) \subseteq \text{dom}(\hat{\rho}) \wedge \forall x \in \text{dom}(\rho) \forall t_x \forall \rho_x : \\ (\rho(x) = \text{close } t_x \text{ in } \rho_x) \Rightarrow (t_x \in \hat{\rho}(x) \wedge \rho_x \mathcal{R} \hat{\rho})$$

- The concrete domain must be included in the abstract one (is an over-approximation)
- For all variables $x \in \text{dom}(\rho)$, if $\rho(x) = \text{close } t_x \text{ in } \rho_x$ then
 - $t_x \in \hat{\rho}(x)$: the function is predicted
 - $\rho_x \mathcal{R} \hat{\rho}$: the captured environment ρ_x is also correct
 - This second condition is recursive: ρ_x may contain closures, whose environments must also be checked

The abstract environment over-approximates every concrete closure, including all environments they recursively capture

Relation between the values

$$v \ V (\hat{\rho}, \hat{v}) \text{ iff } \forall t \ \forall \rho : (v = \text{close } t \text{ in } \rho) \Rightarrow (t \in \hat{v} \wedge \rho \ R \hat{\rho})$$

- The values must agree as well
- For all t , and for all ρ in $(v = \text{close } t \text{ in } \rho)$
 - $t_x \in \hat{v}$: the value is predicted
 - $\rho_x \ R \ \hat{\rho}$: the captured environment ρ_x is also correct.
- This check is recursive: environments may contain closures

A concrete value is a closure, so we must approximate both the function and all the environments it captures

Subject reduction

Theorem

If $\rho \ R \ \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie$ implies $(\hat{C}, \hat{\rho}) \models ie'$

Idea: if there is a possible evaluation of the program such that the function at a call point evaluates to some abstraction, then this abstraction has to be in the set of possible abstractions computed by the analysis

All acceptable analysis results remain acceptable under evaluation

Proof idea

We assume that $\rho \ R \ \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ and prove $(\hat{C}, \hat{\rho}) \models ie'$ by **induction** on the structure of the **inference tree** for $\rho \vdash ie \rightarrow ie'$

For each rule, we show CFA is preserved

Subject reduction

Theorem

If $\rho R \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie$ implies $(\hat{C}, \hat{\rho}) \models ie'$

Proof

We assume that $\rho R \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ and prove $(\hat{C}, \hat{\rho}) \models ie'$ by **induction** on the structure of the **inference tree** for $\rho \vdash ie \rightarrow ie'$

The case **[var]**. Here $\rho \vdash ie \rightarrow ie'$ is $\rho \vdash x^\ell \rightarrow v^\ell$ because $x \in \text{dom}(\rho) \wedge v = \rho(x)$

If $v = c$ there is nothing to prove, so suppose that $v = \text{close } t_0$ in ρ_0 .

From $\rho R \hat{\rho}$ we get $v V (\hat{\rho}, \hat{\rho}(x))$ and hence

$t_0 \in \hat{\rho}(x)$ and $\rho_0 R \hat{\rho}$

From $(\hat{C}, \hat{\rho}) \models ie$ we get

[var] $(\hat{C}, \hat{\rho}) \models x^\ell$ iff $\hat{\rho}(x) \subseteq \hat{C}(\ell)$

$\hat{\rho}(x) \subseteq \hat{C}(\ell)$, and hence $t_0 \in \hat{C}(\ell)$.

Since $t_0 \in \hat{C}(\ell)$ and $\rho_0 R \hat{\rho}$ we have established $(\hat{C}, \hat{\rho}) \models ie'$.

the value is already
predicted by the
analysis

$v V (\hat{\rho}, \hat{v})$ iff $\forall t \forall \rho : (v = \text{close } t \text{ in } \rho) \Rightarrow (t \in \hat{v} \wedge \rho R \hat{\rho})$

Subject reduction

Theorem

If $\rho R \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie$ implies $(\hat{C}, \hat{\rho}) \models ie'$

Proof

The case **[fn]**. Here $\rho \vdash ie \rightarrow ie'$ is $\rho \vdash (\text{fn } x \Rightarrow e_0)^\ell \rightarrow (\text{close } (\text{fn } x \Rightarrow e_0) \text{ in } \rho_0)^\ell$ where $\rho_0 = \rho \upharpoonright FV(\text{fn } x \Rightarrow e_0)$

From $(\hat{C}, \hat{\rho}) \models ie$ we get $\{\text{fn } x \Rightarrow e_0\} \subseteq \hat{C}(\ell)$. **[fn]** $(\hat{C}, \hat{\rho}) \models (\text{fn } x \Rightarrow e_0)^\ell$ iff $\{\text{fn } x \Rightarrow e_0\} \subseteq \hat{C}(\ell)$

From $\rho R \hat{\rho}$ we get we immediately get $\rho_0 R \hat{\rho}$ and then we can establish $(\hat{C}, \hat{\rho}) \models ie'$.

[close] $(\hat{C}, \hat{\rho}) \models (\text{close } t_0 \text{ in } \rho)^\ell$ iff $\{t_0\} \subseteq \hat{C}(\ell) \wedge \rho R \hat{\rho}$

the abstraction is already
recorded in $\hat{C}(\ell)$

Subject reduction

Theorem

If $\rho R \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \vDash ie$ implies $(\hat{C}, \hat{\rho}) \vDash ie'$

Proof

The case [\[app₁\]](#). Here $\rho \vdash ie \rightarrow ie'$ is $\rho \vdash (ie_1 ie_2)^\ell \rightarrow (ie'_1 ie_2)^\ell$ because $\rho \vdash ie_1 \rightarrow ie'_1$

From the induction hypothesis applied to $(\hat{C}, \hat{\rho}) \vDash ie_1$, $\rho R \hat{\rho}$ and $\rho \vdash ie_1 \rightarrow ie'_1$, we get $(\hat{C}, \hat{\rho}) \vDash ie'$.

This suffices to establish $(\hat{C}, \hat{\rho}) \vDash ie'$.

Subject reduction

Theorem

If $\rho R \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie$ implies $(\hat{C}, \hat{\rho}) \models ie'$

Proof

The case **[app_{fn}]**.

Here $\rho \vdash ie \rightarrow ie'$ is $\rho \vdash (\text{close } (\text{fn } x \Rightarrow t_0^{\ell_0}) \text{ in } \rho_1)^{\ell_1} v_2^{\ell_2} \rightarrow (\text{bind } \rho_1[x \mapsto v_2] \text{ in } t_0^{\ell_0})^{\ell}$

From $(\hat{C}, \hat{\rho}) \models ie$ we have $(\hat{C}, \hat{\rho}) \models (\text{close } (\text{fn } x \Rightarrow t_0^{\ell_0}) \text{ in } \rho_1)^{\ell_1}$

which yields $\{\text{fn } x \Rightarrow t_0^{\ell_0}\} \subseteq \hat{C}(\ell_1)$ and $\rho R \hat{\rho}$.

Further, we have $(\hat{C}, \hat{\rho}) \models v_2^{\ell_2}$

[close] $(\hat{C}, \hat{\rho}) \models (\text{close } t_0 \text{ in } \rho)^{\ell}$ iff $\{t_0\} \subseteq \hat{C}(\ell) \wedge \rho R \hat{\rho}$

[case $v_2 = c$]: immediate.

[bind] $(\hat{C}, \hat{\rho}) \models (\text{bind } \rho \text{ in } it_0^{\ell_0})^{\ell}$ iff $(\hat{C}, \hat{\rho}) \models it_0^{\ell_0} \wedge \hat{C}(\ell_0) \subseteq \hat{C}(\ell) \wedge \rho R \hat{\rho}$

[case $v_2 = \text{close } t_2 \text{ in } \rho_2$]: it follows immediately from $(\hat{C}, \hat{\rho}) \models v_2^{\ell_2}$.

Finally, the first universally quantified formula of the definition of $(\hat{C}, \hat{\rho}) \models ie$ gives

$(\hat{C}, \hat{\rho}) \models t_0^{\ell_0}$, $\hat{C}(\ell_2) \subseteq \hat{C}(\ell)$ and $(\rho_1[x \mapsto v_2]) R \hat{\rho}$ leading to establish $(\hat{C}, \hat{\rho}) \models ie'$.

we simulate all possible function calls

Subject reduction

Theorem

If $\rho R \hat{\rho}$ and $\rho \vdash ie \rightarrow ie'$ then $(\hat{C}, \hat{\rho}) \models ie$ implies $(\hat{C}, \hat{\rho}) \models ie'$

Fact

$(\hat{C}, \hat{\rho}) \models it^{\ell_1}$ and $\hat{C}(\ell_1) \subseteq \hat{C}(\ell_2)$ then $(\hat{C}, \hat{\rho}) \models it^{\ell_2}$

Proof

The case **[bind₂]**. Here $\rho \vdash ie \rightarrow ie'$ is $\rho \vdash (\text{bind } \rho_1 \text{ in } v_1^{\ell_1})^\ell \rightarrow v_1^\ell$

From $(\hat{C}, \hat{\rho}) \models ie$ we have $(\hat{C}, \hat{\rho}) \models v_1^{\ell_1}$ and $\hat{C}(\ell_1) \subseteq \hat{C}(\ell)$

The desired $(\hat{C}, \hat{\rho}) \models v_1^\ell$ follows from **Fact**.

This completes the proof. ■

[bind] $(\hat{C}, \hat{\rho}) \models (\text{bind } \rho \text{ in } it_0^{\ell_0})^\ell$ iff $(\hat{C}, \hat{\rho}) \models it_0^{\ell_0} \wedge \hat{C}(\ell_0) \subseteq \hat{C}(\ell) \wedge \rho R \hat{\rho}$

CFA predictions are stable under evaluation

Back to Example 1

$$\begin{array}{c} ((\text{fn } x \Rightarrow x^1)^2 (\text{fn } y \Rightarrow y^3)^4)^5 \\ \downarrow^* \\ (\text{Close}(\text{fn } y \Rightarrow y^3) \text{ in } [])^5 \end{array}$$

$$\begin{array}{c} (\hat{C}, \hat{\rho}) \models ((\text{fn } x \Rightarrow x^1)^2 (\text{fn } y \Rightarrow y^3)^4)^5 \\ \downarrow \text{By Theorem 3.1} \\ (\hat{C}, \hat{\rho}) \models (\text{Close}(\text{fn } y \Rightarrow y^3) \text{ in } [])^5 \end{array}$$

Exercise: Verify it

Back to Example 1

$$(\hat{C}, \hat{\rho}) \models (\text{Close}(\text{fn } y \Rightarrow y^3) \text{ in } [])^5$$

iff

$$\{(\text{fn } y \Rightarrow y^3)\} \subseteq \hat{C}(5) \wedge [] R \hat{\rho}$$

	$(\hat{C}_e, \hat{\rho}_e)$
1	$\{\text{fn } y \Rightarrow y^3\}$
2	$\{\text{fn } x \Rightarrow x^1\}$
3	$\emptyset$
4	$\{\text{fn } y \Rightarrow y^3\}$
5	$\{\text{fn } y \Rightarrow y^3\}$
x	$\{\text{fn } y \Rightarrow y^3\}$
y	$\emptyset$

✓

trivially true

$$\rho \mathcal{R} \hat{\rho} \text{ iff } \text{dom}(\rho) \subseteq \text{dom}(\hat{\rho}) \wedge \forall x \in \text{dom}(\rho) \forall t_x \forall \rho_x : \\ (\rho(x) = \text{close } t_x \text{ in } \rho_x) \Rightarrow (t_x \in \hat{\rho}(x) \wedge \rho_x \mathcal{R} \hat{\rho})$$

$$[\text{close}] (\hat{C}, \hat{\rho}) \models (\text{close } t_0 \text{ in } \rho)^\ell \text{ iff } \{t_0\} \subseteq \hat{C}(\ell) \wedge \rho \mathcal{R} \hat{\rho}$$